

Lai Jiaqi

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RESEARCH INTERESTS

Topics: Human–AI Trust Dynamics; Algorithmic Delegation; Autonomous AI Systems & Society.

Methods: Self-Evolving Multi-Agent Systems; Agent-Based Modelling; Coupled Dynamical Systems Modelling (Hawkes Process, Friedkin–Johnsen); Bayesian Truth Serum; Mixed-Methods Behavioural Experiments.

EDUCATION

Nanyang Technological University | Singapore

Aug. 2024 – Jul. 2025

MSc Information Studies

- Relevant Courses: Machine Learning Implementation; Disinformation & Information Professions (A, 5/5); Text & Web Mining; Critical Inquiry about Information Warfare and Computational Propaganda (A, 5/5).
- Enrolled in **PhD-level Mathematical Economics by special permission** (A, 5/5).

University College London | London, UK

Sep. 2021 – Jun. 2024

BSc Psychology & Language Sciences

- Relevant Courses: Deep Learning & Language Processing (1st Class); R & Statistical Methods (1st Class); Human-Computer Interaction; Cognitive Neuroscience; Behavioural Psychology.

PUBLICATIONS & SELECTED ACADEMIC WRITINGS

Peer-Reviewed Publication

Steed, A., & Lai, J. (2025). Comparison of hand tracking-based and controller-based interaction in a consumer virtual reality game. *Virtual Reality*, 29(3), 120. [JCR Q1]

Working Papers

Lai, J., Hou, L., & Huang, W. (2025). “Stability of AI Governance Systems: A Coupled Dynamics Model of Public Trust and Social Disruptions.” *Manuscript in preparation* — Submitted to *AAAI/ACM AIES 2026*.

Derives closed-form spectral stability conditions ($\rho(J_{2n}) < 1$) for a bidirectionally coupled Hawkes–FJ trust dynamics system; demonstrates irreversible governance collapse under echo-chamber topologies.

Selected Academic Writing

Lai, J., & Dechant, M. (2024). “Can freedom lead to trust & satisfy? How user autonomy influences trust perception in LLM-powered conversational agents.” *Manuscript in preparation*.

Mixed-methods study ($N = 86$) challenging standard assumptions about user autonomy and trust in human–AI interaction; identified latent-variable measurement limitations.

Lai, J., Wang, T. and Li, X. (2024). “The Disinformation Campaigns about the Israel-Palestine Conflict in the Digital Era.” *MSc Critical Inquiry Project*. [Awarded Full Marks] — available as a writing sample.

Analysed the systemic impact of generative AI on state-sponsored computational propaganda through the lens of Strategic Narrative Warfare theory.

RESEARCH & POLICY EXPERIENCE

AI Singapore | Singapore

Apr. 2026 – present

Research Assistant, under Dr. William Tjhi

- Building a **self-evolution agentic framework** that iteratively refines its own policy rules, deployed inside a **policy simulation sandbox** for stress-testing adaptive regulation on the Barcelona homelessness problem.
- Driving the evolution loop with **competing conceptual frameworks**: instantiating agent personas that embody distinct normative schools (e.g. utilitarian, egalitarian) and **debate** simulated policy outcomes, so the framework’s rules evolve through structured moral argumentation.
- Validating evolved policies via an **Agent-Based Modelling (ABM)** section—simulating regulator–developer–public interaction to assess how the adapted governance shapes long-run policy outcomes.
- Preparing a manuscript on self-evolving agents for policy evaluation, **targeting AAAI (July 2026)**.

NTU School of Economics | Singapore

Jan. 2025 – Jul. 2025

Independent Research Project, PhD-level Mathematical Economics

- Constructed a **discrete-time coupled dynamical system** (**Hawkes process** $\times$ **Friedkin–Johnsen dynamics**) formally characterising bidirectional feedback between AI controversy events and public trust evolution.
- Derived **Jacobian-based spectral stability conditions** ($\rho(J_{2n}) < 1$), establishing tipping-point thresholds for trust collapse across heterogeneous network topologies.
- Applied **mechanism design** tools (incentive-compatible elicitation) to model AI governance under asymmetric information; evaluated how institutional design shapes long-run collective trust trajectories.

UCL Interaction Centre (UCLIC) | *London, UK*

Sep. 2022 – Apr. 2024

Research Assistant, under Prof. Yvonne Rogers (UCLIC Director)

- Designed and executed a **mixed-methods within-subject experiment** ($N = 86$) comparing LLM-powered and branching-dialogue agents, examining how user autonomy influences trust and satisfaction perceptions.
- Employed **non-parametric inference** and **moderated linear regression** to test autonomy–trust–satisfaction pathways; complemented quantitative results with **thematic analysis** of open-ended responses.
- Key finding: higher perceived autonomy in LLM-based interaction did not increase trust or satisfaction, suggesting users prioritise **problem-solving capability** over interaction freedom—informing future work on CFA/SEM measurement of latent trust constructs.

NTU WKWSCI | *Singapore*

Sep. 2024 – Apr. 2025

Machine Learning Researcher

- Analysed cross-cultural AI sentiment dynamics on Weibo and X/Twitter using **NLP pipelines** (VADER, NRCLEX, BERTopic, PageRank) to characterise opinion propagation across two governance contexts.
- Trained and benchmarked **transformer and recurrent classifiers** (BERT, GRU, LSTM) for cross-lingual sentiment classification under distribution shift.

UCL Computer Science | *London, UK*

Oct. 2023 – May. 2024

Research Assistant

- Designed a **controlled VR study** on human behavioural responses to interaction modalities; contributed to statistical analysis and manuscript writing — *published in Virtual Reality*.

China Development Research Foundation | *Beijing, China*

Oct. 2024 – Mar. 2025

Policy Research Assistant

- Evaluated **AI deployment gaps** between policy intent and real-world adoption in education and rural contexts; identified structural barriers and **incentive misalignments** shaping local governance outcomes.
- Synthesised cross-disciplinary evidence into **policy evaluation reports** for senior stakeholders; defined success indicators and risk frameworks for national AI governance programmes.

INDUSTRY EXPERIENCE

Golden Gate Ventures | *Singapore*

Jun. 2025 – Aug. 2025

Machine Learning Intern

- Deployed a **multi-agent AI pipeline** over heterogeneous data sources, reducing research turnaround.
- Fine-tuned **BERT** and **Qwen** for domain-specific classification; gained first-hand exposure to **production-scale AI deployment** decisions (model selection, output alignment, stakeholder trust calibration).

Canopy | *Singapore*

Aug. 2025 – Jan. 2026

Data & Sales Enablement Executive

- Developed a **quantitative pricing model** (logistic regression, GAM) through iterative coordination with operational teams; coordinated cross-departmental data integration for investor due diligence.
- Gained practical exposure to **incentive structure design** in organisational decision-making contexts.

SKILLS

Formal & Quantitative Methods: Self-Evolving Multi-Agent Systems, Agent-Based Modelling, Coupled Dynamical Systems, Hawkes Process, Mechanism Design, Bayesian Truth Serum, MCMC, Bayesian Inference, Game Theory.

Machine Learning & Programming: Python, TensorFlow, BERT/Qwen (fine-tuning), GRU/LSTM/RNN, NLP (VADER, BERTopic, NRCLEX), CrewAI/LangChain (multi-agent pipelines).